

(1) Define IOT and its characteristics.

IOT : The Internet of Things refer to a network of physical objects (things) embedded with sensors, software & electronics plus network connectivity that enables them to collect the process, exchange the data with other devices and systems over the internet without requiring human interaction.

Key characteristics of IOT

- ① **connectivity** : connects hardware devices & objects across the wired or wireless networks
- ② **Dynamic & Adaptive** : continuously adapts to changing operational environments and contexts automatically
- ③ **Self-configuring** : devices can configure themselves, update firmware, & integrate to network with minimal interaction
- ④ **Interoperability** : enables communication across varied hardware architectures as platforms & protocols.

⑤ Unique Identification: Every connected device is uniquely identifiable using IP address, Mac address & UUID.

⑥ Data Intelligence & Automation
↳ Transforming raw sensor readings into actionable insights & automated workflow.

⑦ Scalability → capable of handling scaling and/or nodes efficiently across global infra.

(2) Explain the Evolution of IIoT from M2M communication?

⇒ Machine to Machine communication: Traditional M2M represents point to point communication between dedicated hardware devices and remote monitoring stations, usually over cellular or proprietary wired network in isolated silos.

Evolutionary Progression to IIoT:

① P2P Hardware Telemetry: closed single purpose monitoring setups relying on specialized hardware &

② IP Protocol integration;

↳ Adaptation of standard 32 stack onto microcontroller unit, enabling seamless internet based access.

③ cloud computing & Big Data

↳ centralization of Data processing storage & analytics from on-premise servers to scalable cloud platform

④ Open Ecosystem & standardization

↳ Adoption of standardized lightweight web protocols (MQTT, CoAP, REST API) enable seamless interoperability across heterogeneous platform & multi vendor devices

⑤ Draw and Explain the Four layer IoT architecture?

Data Flow / Control Flow

APPLICATION Layer
(Smart app & management)

Data processing
(Data processing, Data Analytics,
Decisions)

Network Layer
(Internet Gateway, Network Gateway,
Network Tech (Data Acquisition System))

Sensing Layer
(Physical objects, sensors &
actuators)

[Diagram drawn
from image source
in tellichath]

(3)

The standard four layer IoT architecture organizes data processing from physical hardware acquisition up to end user app:

④ Application Layer

- ↳ Delivery specialized content aware user interfaces, dashboards, and domain logic
- ↳ Key Technologies & component
 - ↳ smart home apps, industrial monitoring, fleet tracking, smart health

③ Data processing Layer

- ↳ collect, stores, processes & analyzes high volume sensor stream; manages security

② Network Layer

- ↳ Routes & transmits collected sensor data securely to processing hubs or clouds

① Sensing Layer: Detect physical parameters in the environment & convert them to digital signal

④ ~~Diff~~ Differential between Sensing Layer and network layer

⇒ Sensing Layer Network Layer

① Data acquisition from physical environment & mechanical actuation

① Data transmission packet routings - secure network bridging

② Sensors, actuators, RFID chips, micro controllers

② Routers, gateways, modems, accesspoint, switches

③ Raw analog signals or digital register values

③ Structured IP network packet, frames, socket, streams

④ I2C, SPI, UART, RFID, NFC, oneWire

④ IPV4, IPV6, WIFI, LoRaWAN, TCP/IP, MQTT, CoAP

⑤ Layer 1 (Bottom sensing physical Tier)

⑤ Layer 2 (Transmission connectivity Tier)

Q Explain the Functions of Data processing and application layers

→ Functions of Data processing (middleware) Layer

1) Data ingestion & storage

↳ stores massive streams of time series data using scalable SQL/Non-SQL db

2) Filtering & Aggregation

↳ Eliminates duplicate readings
filters out signal noise.
normalizes incoming data traffic

3) Analytics & Intelligence

↳ Performs realtime event processing
machine learning inference &
threshold checks

4) Device check & identity management

↳ manages authentication, firmware
provisioning & health telemetry
connects nodes

Functions of Application Layer

- ↳ User Presentation
- ↳ Business Logic Execution
- ↳ Enterprise API Integration

⑥ Explain communication API used in IOT

⇒ IOT system primarily rely on two major architectural patterns of API communication:

① REST API (Representational State Transfer)

↳ model → Request-Response architecture over HTTP or HTTPS protocols

↳ characteristics → Stateless, Highly Scalable, uses standard HTTP verbs (GET, POST, PUT, DELETE)

→ Use case → Fetching device status, configuration, system settings and querying stored historical sensor metrics

③

② websocket based communication

↳ mode -> full duplex continuous bi directional socket connection over a single TCP connection

↳ characteristics : Low communication overhead, real time message delivery, event driven

↳ use case : Real time ~~chore~~ telematics, continuous sensor streaming, live video/audio feeds, and instantaneous remote actuation.

① compare IOT & M2M communication

M2M (Machine to Machine)

↳ direct point to point connection between isolated hardware machines

↳ Proprietary Protocols or specialized Protocols ^{cellular}

↳ limited scalability - expansion requires machine or departmental size

↳ Device to device or device to Server ~~server~~ connection type

IoT (Internet of Things)

- Broad, interconnected network leveraging cloud, edge & IP infra
- Protocols: open, standardized IP & web protocols (HTTP, MQTT)
- Virtually unlimited scalability due to elastic cloud hosting & ~~open~~ open standards
- Data shared across cloud platforms, enterprise tools, and analytics engines
- Connection type:
 - ↳ Device to Device, Device to cloud, cloud to cloud, Device to user

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(8) Explain security challenges in IOT

↳ Resource constraint : Embedded micro controller lack the CPU power and memory required for computationally heavy encryption standards.

↳ Weak default Credentials : Many consumer IOT devices ship with hard coded default passwords, making them easy targets for automated botnet attacks.

↳ Lack of firmware update mechanisms : Inexpensive sensor often lack over the Air update ~~info~~ infrastructure, making long term patching difficult or impossible.

⇒ Unencrypted Network Traffic

↳ Means mitling unencrypted telemetry data over public ~~wifi~~ Wi-Fi, cellular links exposes ~~sys~~ system to ~~easy~~ eavesdropping and man in the middle attacks.

↳ Physical tampering.

(9) Discuss deployment challenges in IOT

↳ Interoperability & Standards

↳ Integrating components across varied vendors legacy protocols and hardware architectures create severe system fragmentation

↳ Power and Battery management

↳ Designing ultra low power embedded solutions capable of operating for years on small batteries in field conditions

↳ Reliable connectivity

↳ maintaining continuous wireless connectivity across remote, industrial, underground, or outdoor environments.

↳ Data ingestion Scalability

↳ Ingesting, processing and indexing massive continuously streaming time series data without service degradation

10 Explain any three real world applications of IIOT

① Smart Homes

↳ Interconnect Household devices such as smart thermostat, lighting system, security camera, and voice assistant over Wi-Fi. Enables automated energy management, automated scheduling & remote monitoring via mobile application.

② Industrial IIOT & Predictive maintenance

↳ Embed vibration, temperature & acoustic sensors into factory machinery. By monitoring real time operating metrics, analytics engine predict mechanical wear and component failure prior to breakdown ~~and~~ avoiding costly downtime.

③ Smart Agriculture

↳ deploys wireless soil sensors measuring moisture, pH, and ambient temperature alongside

automated drip irrigation system
water and fertilizer are delivered
precisely when needed, conserving
natural resources while
maximizing crop yields.